

Validity on Portfolio Allocation Data-Based Models

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Objective

The objective of this talk is to provide attendees with a more realistic view of the biases to which they are exposed as portfolio managers and to show how failing to account for these biases can significantly affect the real performance of their portfolio.

- Blind construction of an investment portfolio
- Out-of-sample result of the blind construction of an investment portfolio
- How to more accurately simulate the reality of a portfolio manager
- Typical biases
- Look-ahead bias and overfitting
- “Walk-Forward” evaluation framework
- Survivorship bias
- Liquidity biases

Agenda

- Traded volume
- Transaction costs
- The limit order book
- Execution cost
- Sample window bias and manager overfitting
- Comparison across windows: The illusion of results
- Examples of methodologies to robustly validate a Sharpe ratio
- Proposed framework to evaluate a portfolio
- Other common issues when validating a portfolio

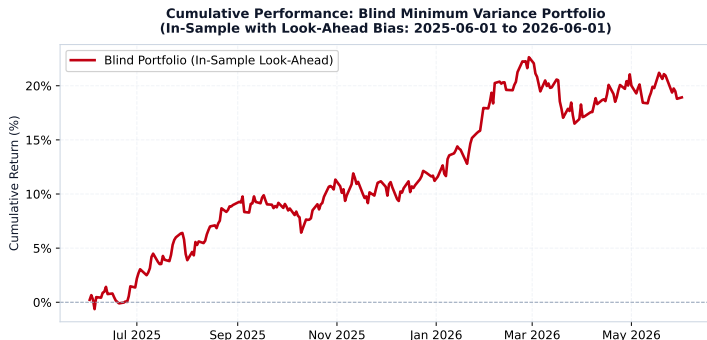
Blind construction of an investment portfolio

Let's go back in time for a moment and assume that we are on June 01, 2026. We want to construct a portfolio with the following stocks: **NVDA** (Nvidia), **PG** (Procter & Gamble), **LLY** (Eli Lilly), **MCD** (McDonald's), **CVX** (Chevron), **CAT** (Caterpillar), **AMZN** (Amazon), **JPM** (JPMorgan Chase), **GOOGL** (Google), and **UNH** (UnitedHealth).

Would you consider it a diversified portfolio?

Blind construction of an investment portfolio

Now, we only want to minimize the risk of the portfolio; for this, we will use the minimum variance portfolio of Markowitz (1952) using data from the past year (12 months). As of June 01, 2026, the cumulative return plot if that money had been invested on June 01, 2025 is the following:



Blind construction of an investment portfolio

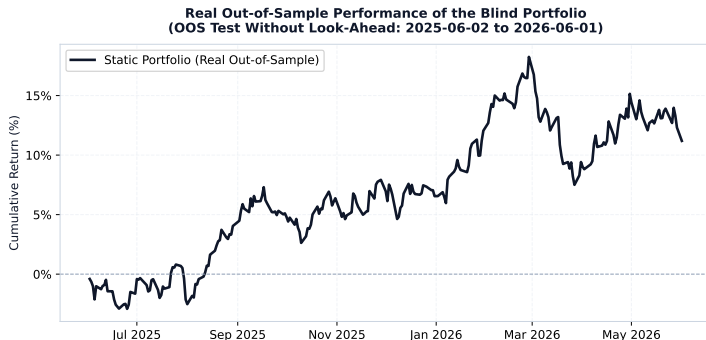
The performance metrics of this portfolio are the following:

Metric	Value
Annual volatility	8.85%
Total return	18.91%
Annualized return	18.99%
Sharpe ratio	2.15
Maximum Drawdown	-5.01%

Table: Basic performance metrics of the blind portfolio (In-Sample with Look-Ahead)

Out-of-sample result of the blind construction of an investment portfolio

What would have happened if a manager on June 01, 2025 had constructed that portfolio without looking into the future (using prior data from 2024 to 2025) and kept it fixed during that year? The out-of-sample plot is:



Out-of-sample result of the blind construction of an investment portfolio

When evaluating that portfolio honestly out-of-sample between June 01, 2025 and June 01, 2026, the real performance metrics are the following:

Metric	Value
Annual volatility	10.78%
Total return	11.15%
Annualized return	11.20%
Sharpe ratio	1.04
Maximum Drawdown	-9.09%

Table: Performance metrics of the blind portfolio out-of-sample (Static OOS: 2025–2026)

Out-of-sample result of the blind construction of an investment portfolio

Those metrics cannot be replicated!, what reasons do you think invalidate the performance that had been seen for the portfolio?

On top of everything, this model is also not taking into account transaction costs, **which would turn into a greater loss for the portfolio manager**, but we will see that later.

How to more accurately simulate the reality of a portfolio manager

What we just saw is the result of blindly trusting a portfolio construction model; in academic settings, falling into these types of problems is valid and works very well to learn the difference between theory and reality. However, when managing a real portfolio, blindly trusting these types of models can lead to **significant losses of value**, and that is precisely why it is important to **simulate the reality of a portfolio manager**.

Although this talk does not cover all the realities that a portfolio manager faces, **the objective of this talk is to provide a framework closer to reality when constructing an investment portfolio**.

How to more accurately simulate the reality of a portfolio manager

As we saw a moment ago, a manager faces the need to validate their investment portfolio construction strategies in the past; however, it is important to take into account other very common situations. To exemplify one of these situations, let's go back to June 01, 2024 for a moment: was Canacol Energy listed on the stock exchange? As of today, is Canacol Energy still listed on the stock exchange?

This and other situations will be addressed later.

Typical biases

- Look-ahead bias
- Survivorship bias
- Liquidity bias
- Overfitting bias (Mentioned in this talk)
- Sample window bias (Mentioned in this talk)
- Adverse selection bias (Not covered in this talk)

Look-ahead bias

So far we saw that the initial portfolio seemed perfect (Return of 18.91% and Sharpe of 2.15), but when evaluating it honestly out-of-sample the Sharpe dropped to 1.04 and the return to 11.15%. Why did this happen?

In the first model, we optimized with the data from 2025 to 2026 to evaluate that same period: a manager on June 01, 2025 would not have known them!

That is called *look-ahead bias*

Look-ahead bias

This bias formally appears when information from the future is used to calibrate models in the past, producing inflated metrics that do not replicate in the real market:

The windows of the model with *Look-Ahead Bias* were:

Training: 01-06-2025 $\rightarrow$ 01-06-2026 (*Sharpe 2.15*)

Testing: 01-06-2025 $\rightarrow$ 01-06-2026

While the out-of-sample model used:

Training: 01-06-2024 $\rightarrow$ 01-06-2025

Testing (OOS): 02-06-2025 $\rightarrow$ 01-06-2026 (*Sharpe 1.04*)

This notation will be the basis for **The “Walk-Forward” evaluation framework**

“Walk-Forward” evaluation framework

The “Walk-Forward” evaluation framework, originally formalized by Pardo (1992), is an iterative process where, in principle, an initial training window and a testing window are established over the known data; the model is tested on that testing window, and performance metrics are recorded. In the next step, that testing window becomes part of the training data and the model is tested on a new window. Following the example, a way to start the evaluation process using “Walk-Forward” is as follows.

For the first step:

Training: 01-06-2024 → 01-06-2025

Testing: 02-06-2025 → 01-09-2025

“Walk-Forward” evaluation framework

The portfolio performance metrics are computed, and now that testing window is added to the new training data, establishing a new testing window that covers the same amount of time so that they are comparable.

For the second step:

Training: 01-06-2024 → 01-09-2025

Testing: 02-09-2025 → 01-12-2025

“Walk-Forward” evaluation framework

Again, performance metrics are computed and the process continues until reaching the last testing window, that is, the one ending on the current day (in this case, since we went back in time, it would be June 01, 2026).

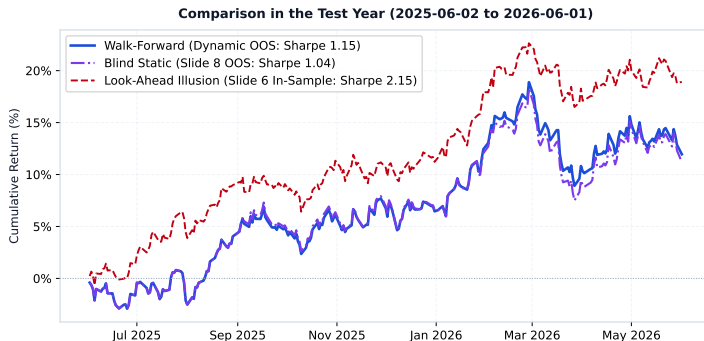
The last step of this process would have the following training and testing windows:

Training: 01-06-2024 → 01-03-2026

Testing: 02-03-2026 → 01-06-2026

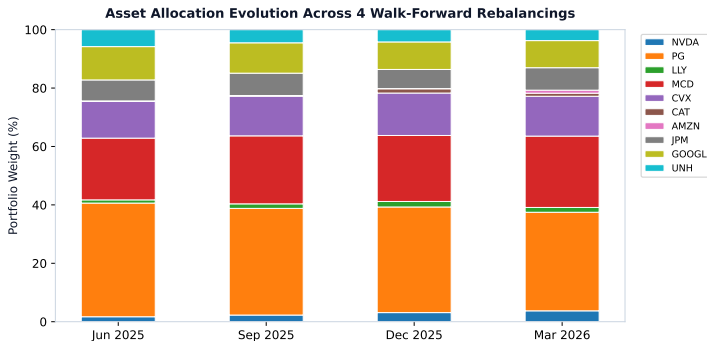
“Walk-Forward” evaluation framework

Cumulative return in the test year (2025-06 to 2026-06): Dynamic Walk-Forward versus the blind model and the static model:



“Walk-Forward” evaluation framework

Asset allocation evolution across the 4 quarterly rebalancings:



“Walk-Forward” evaluation framework

Applying this process to the portfolio we already had, the portfolio performance metrics would be the following:

Metric	Value
Annual volatility	10.43%
Total return	11.93%
Annualized return	11.98%
Sharpe ratio	1.15
Maximum Drawdown	-8.39%
Average turnover ratio	4.26%

Table: Basic performance metrics of the portfolio under “Walk-Forward” (1 Year)

“Walk-Forward” evaluation framework

Comparing the evaluation metrics in the test year (02-06-2025 to 01-06-2026):

Metric	WF (Dynamic)	Static OOS	Look-Ahead
Annual volatility	10.43%	10.78%	8.85%
Total return	11.93%	11.15%	18.91%
Annualized return	11.98%	11.20%	18.99%
Sharpe ratio	1.15	1.04	2.15
Maximum Drawdown	-8.39%	-9.09%	-5.01%
Average turnover ratio	4.26%	0.00%	NA

Table: Comparison in the test year (2025–2026): Dynamic vs Static vs Look-Ahead

The metrics are much more realistic!

Survivorship bias

We have already covered look-ahead bias with our first theoretical portfolio; now we are going to focus on another of the most common biases: **Survivorship bias**.

To illustrate this bias, let's return to the Canacol Energy example from a few slides ago: as of today, Canacol Energy is no longer listed on the stock exchange; however, 2 years ago it was still part of the available investment universe. An investor 2 years ago could have considered it an investment opportunity, therefore **it is not correct to eliminate this stock in our portfolio construction models if it was still trading in the data we have**.

Survivorship bias

This bias is very important and is rarely taken into account in portfolio construction research; the most common practice is to eliminate from the investment universe stocks that do not have complete data in the window used to build the model, **and that exact behavior is what creates the bias!**. Right now, it is likely that almost no model would remove Nvidia from its portfolio, but **what happens if Nvidia goes bankrupt in 2 years? Would the manager really have removed Nvidia from their portfolio?** (*I am not saying it will go bankrupt, this is merely illustrative*)

Eliminating stocks from the investment universe because in the historical data we have they cease to exist is also equivalent to looking ahead. **The manager does not know it!**

There are many methodologies used to handle this bias; however, the one focused purely on the available data consists of **preserving the data of delisted stocks and dynamically adjusting the investment universe** to handle these types of events. If stocks are delisted it does not necessarily imply they went bankrupt; the liquidation value of those shares could be estimated, although that is not something we will cover in this section. *If you wish to review one of the models used to model delisting, you can read the model proposed by Shumway (1997).*

Survivorship bias

The other form of survivorship bias occurs when the investment universe expands within the data window at hand. For example, if I had data from the US market from 2 years ago until today, do you know of any stock that has been listed on the US market in these 2 years? **Would you deliberately remove it from your portfolio because it did not exist before?**

To give a well-known example, *SpaceX* went public on June 12, 2026: **would you never have considered it for your investment portfolio? Do you think managers at JPMorgan, Goldman Sachs, or Morgan Stanley did not take it into account?.**

For these types of situations, it is also useful to dynamically adjust the portfolio manager's investment universe in our model testing.

The other most common biases consist of not taking into account transaction costs and potential liquidity issues related to the assets traded in the portfolio. The liquidity biases that we will cover in this talk are the following:

- Traded volume
- Transaction costs
- Execution cost

Traded volume

Traded volume is the **number of shares that change ownership throughout a market session**. It is independent of price; that is, if 100 shares of a company were traded at a price of \$100 and 100 shares of the same company at \$105 in that same session, the traded volume is 200 shares.

The other volume concept is **dollar volume**, which is measured in currency units. Following the previous example, the dollar volume of the shares of those companies would be \$20,500.

Traded volume

So far we know the concepts of traded volume and dollar volume, but why are these concepts important to take into account when testing portfolio construction models?

Suppose that today I have to invest 100 million USD in the ADR of Grupo Cibest, the traded volume for the day is 500,000 shares, and its dollar volume is 50 million USD: **could I really invest the 100 million dollars in the ADR of Grupo Cibest?**

Failing to take into account this and the issues discussed later corresponds to a model that assumes **a frictionless market**

Transaction costs

Agents in the market can rarely directly buy or sell the assets belonging to their portfolio; in order to execute these buy/sell operations they rely on intermediaries known as **“brokers”**. They are responsible for matching trades between agents intending to sell and agents intending to buy. This work is not free; they must find counterparties to match orders and therefore charge a commission. Sometimes it is a fixed commission, sometimes it varies and is not the same across all brokers available in the market; if these commissions are not taken into account when testing portfolio construction models, one would be providing information on an unrealistic portfolio and an unreplicable return.

**This is why the average turnover ratio we saw in the
“Walk-Forward” validation is important!**

Transaction costs

Other important transaction costs not covered in this session may include the following:

- Interest rates: Appears especially in leveraged operations and represents the charge for borrowing money to trade
- Taxes: Depending on the country, there may be capital gains taxes on stock market investments
- Custody fees: The fee charged by a financial institution to keep your assets in custody. Very common in futures markets.
- Slippage: The difference between the expected execution price of a market order and the price at which it actually executes.

The last liquidity bias covered in this talk will be execution cost; however, to discuss execution cost it is important to first discuss what a limit order book is.

The limit order book

Electronic markets operate under a limit order book, in which intentions to buy and sell an asset are declared, showing the price and the quantities associated with each. A hypothetical example of what an order book looks like:

Quantity	Bid		Offer	Quantity
		100	110	50
			105	20
			101	100
25	99			
112	98			
53	95			

Table: Example of what a limit order book looks like

The limit order book

In the example book, the numbers on the *bid* side represent buying intentions, while the numbers on the *offer* side show the selling intentions of different market participants. The number between the two is known as the *mid-price*, which is an **average between the best ask price (the lowest) and the best bid price (the highest)**. In an order book there are agents operating simultaneously; first we will talk about *makers*: These agents place limit orders in the market and essentially build the book. The actions a *maker* can take are:

- Place buy orders
- Place sell orders
- Cancel buy orders
- Cancel sell orders

The limit order book

For example, when adding a new limit buy order at a price of **99** and a quantity of **23**, the book updates to the following state:

Quantity	Bid		Offer	Quantity
		100	110	50
			105	20
			101	100
48	99			
112	98			
53	95			

Table: Example of what a limit order book looks like after adding a new buy order

The limit order book

Upon cancelling a sell order at a price of **101** and quantity of **100**, the book updates to the following state:

Quantity	Bid		Offer	Quantity
		102	110	50
			105	20
48	99			
112	98			
53	95			

Table: Example of what a limit order book looks like after cancelling an existing sell order

If the best bid or best ask updates, the mid-price updates!

The limit order book

The other agents are *takers*; they execute “market orders”, which consist of buying or selling a specific quantity determined by the *taker* at whatever price is available in the market. For example, a *taker* wants to buy 15 shares in the last order book seen; the execution price would be **105** and the quantity sold to the *taker* would be 15 shares. Therefore, the book would update to the following state:

Quantity	Bid		Offer	Quantity
			110	50
			105	5
		102		
48	99			
112	98			
53	95			

The limit order book

Now let's go back a moment before the purchase and assume they do not want to buy 15 shares but 40; the best ask is no longer capable of covering the quantity of shares the *taker* desires. Therefore, they must buy the remaining shares at the next best price, namely **110**, making their purchase much more expensive. The book would update to the following state:

Quantity	Bid		Offer	Quantity
		104.5	110	30
48	99			
112	98			
53	95			

This is what is called execution cost!

Execution cost

In the example we saw, the *taker* bought \$8 above the *mid-price*, which is often considered the “equilibrium price” of the market. Relative to the old “equilibrium price”, the *taker* bought approximately 7.84% more expensive, whereas relative to the new mid-price, the *taker* bought approximately 5.26% more expensive.

Furthermore, if the *taker* wishes to immediately exit and sell those shares, the *taker* would have to sell them at a price of 99, leading to a 10% loss of the invested value. This would also be an execution cost that needs to be modeled; in very illiquid assets, the best bid and the best offer can be very far apart, forcing one to incur higher execution costs.

Furthermore, can assets always be bought at their closing price?

If you wish to review models to handle transaction costs more properly, you can consult the following articles:

- Engle, Ferstenberg, and Russell (2006)
- Bertsimas and Lo (1998)

Sample window bias and manager overfitting

At the beginning of the session we mentioned that *overfitting* and *sample window bias* would not be covered in detail; however, they are among the most common dangers in practice.

What happens when a manager tests different evaluation windows?

- The manager tests several time windows (1 year, 1.5 years, 2 years) until finding one where the portfolio performance validates the model used.
- This is known as **Meta-Overfitting** or *Data Snooping*: the model does not truly generate more value, only the window where the market favored the model was selected. **Market conditions rarely repeat such that this performance can be replicated**

Comparison across windows: The illusion of results

The same minimum variance model evaluated under different configurations:

Configuration / Window	Return	Volatility	Sharpe
In-Sample (Look-Ahead 1 year: 2025–2026)	18.91%	8.85%	2.15
Walk-Forward (1 year: 2025–2026)	11.93%	10.43%	1.15
Blind Static (1 year OOS: 2025–2026)	11.15%	10.78%	1.04
Walk-Forward (1.5 years: 2024–2026)	7.46%	13.08%	0.38
Real Static OOS (prior 2 years)	0.54%	13.43%	0.03
Alternative In-Sample (2 years: 2024–2026)	23.00%	11.77%	0.93

Table: Performance sensitivity according to the sample window (*annualized)

The same model can seem like a failure or a good model depending on the chosen window!

Examples of methodologies to robustly validate a Sharpe ratio

To evaluate the robustness of the Sharpe ratio after testing the same model across different time windows, the quantitative literature proposes statistical adjustment frameworks:

- **Deflated Sharpe Ratio (DSR):** Corrects the observed Sharpe by penalizing for the number of trials or variants tested (N), the skewness and kurtosis of returns, and the length of the sample horizon.
- **Probabilistic Sharpe Ratio (PSR):** Calculates the formal probability that the true population Sharpe exceeds a reference benchmark.

Examples of methodologies to robustly validate a Sharpe ratio

Articles on overfitting and Sharpe validation:

- DSR: Bailey and López de Prado (2014)
- PSR: Bailey et al. (2014)

There are more methodologies in the literature to address the biases mentioned in this talk; **whenever working with data-driven models, it is important to keep these biases in mind!**

Proposed framework to evaluate a portfolio construction model

As you could notice, just discussing these biases takes quite some time; unfortunately, **implementing them correctly takes even more time**. Therefore, a framework is usually proposed that validates the portfolio construction model and its benefit relative to the benchmark (*alpha*) in stages:

- Step 1: Build the allocation model
- Step 2: Test the model inside the training sample → If *alpha* exists, continue; otherwise, the strategy is invalidated

Proposed framework to evaluate a portfolio construction model

- Step 3: Test the model under the “Walk-Forward” validation framework → If *alpha* exists, continue; otherwise, the strategy is invalidated
- Step 4: Use asset turnover in the portfolio to apply transaction costs → If *alpha* exists after correcting for transaction costs, continue; otherwise, the strategy is invalidated
- Step 5: Add execution cost modeling → If *alpha* exists after correcting for execution costs, continue; otherwise, the strategy is invalidated

To this list, the steps corresponding to managing the other biases and issues not covered in this talk must be added to continue with the model's validity

Other common issues when validating a portfolio

- *Regime-Switching*: The model's performance can be very good in a particular market regime (bullish, bearish, sideways); however, it is important to test its performance in other market regimes since **we do not know what direction the market will take in the future**
- Lack of *Stress-Testing*: The performance of portfolio construction models can be very good under regular market conditions; however, **what happens to the model under crisis conditions?**

Although these are not issues covered in this talk, they are very relevant for professional portfolio managers, as they can better adjust their risk estimates and the robustness of their performance metrics across different market conditions.

- [1] David H. Bailey and Marcos López de Prado. “The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality”. In: *The Journal of Portfolio Management* 40.5 (2014), pp. 94–107. ISSN: 0095-4918. DOI: 10.3905/jpm.2014.40.5.094. URL: <https://doi.org/10.3905/jpm.2014.40.5.094>.
- [2] David H. Bailey et al. “Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance”. In: *Notices of the American Mathematical Society* 61.5 (2014), pp. 458–471.
- [3] Dimitris Bertsimas and Andrew W. Lo. “Optimal control of execution costs”. In: *Journal of Financial Markets* 1.1 (Apr. 1998), pp. 1–50. ISSN: 13864181. DOI: 10.1016/S1386-4181(97)00012-8. URL: <https://linkinghub.elsevier.com/retrieve/pii/S1386418197000128> (visited on 09/08/2026).

References II

- [4] Robert F. Engle, Robert Ferstenberg, and Jeffrey R. Russell. *Measuring and Modeling Execution Cost and Risk*. Rochester, NY, Apr. 1, 2006. URL: <https://papers.ssrn.com/abstract=1293666> (visited on 09/08/2026).
- [5] Harry Markowitz. "Portfolio Selection". In: *The Journal of Finance* 7.1 (1952), pp. 77–91. ISSN: 0022-1082. DOI: 10.2307/2975974. URL: <https://www.jstor.org/stable/2975974> (visited on 09/08/2026).
- [6] Robert Pardo. *Design, Testing, and Optimization of Trading Systems*. New York: John Wiley & Sons, 1992. ISBN: 978-0471554462.
- [7] Tyler Shumway. "The Delisting Bias in CRSP Data". In: *The Journal of Finance* 52.1 (1997), pp. 327–340. ISSN: 00221082, 15406261. URL: <http://www.jstor.org/stable/2329566> (visited on 09/07/2026).

Questions?

Thank you very much